INCH-POUND

CCC-C-439G
May 15, 1990
SUPERSEDING
CCC-C-439F
August 1, 1973

FEDERAL SPECIFICATION

CLOTH, BUNTING, COTTON, MERCERIZED

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of cotton bunting.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specification

PPP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

Federal Standards

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 8305

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Federal Regulations

Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580.)

2 2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Color Association of the United States

Standard Color Card of America

(Color cards may be available from the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

American Association of Textile Chemists and Colorists (AATCC)

Chromatic Transference scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, 1 Davis Dr., P.O. Box 12215, Research Triangle Park NC 27709-2215.)

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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Sample. When a standard sample is available, the cloth shall match the sample for shade and shall be equal to or better than the sample with respect to all characteristics for which the sample is referenced (see 6.4).

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Yarn. The yarn shall be made from cotton which has been carded, drawn, spun and twisted into two-ply yarn for both warp and filling.

3.4 Color. The color of the finished cloth shall be as specified (see 6.2).

3.4.1 Matching. The color of the finished cloth, shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K. When no standard sample is available the finished cloth shall match the applicable color card (see 2.2), either by cable number or names.

3.4.2 Colorfastness. The finished cloth shall show fastness to light, water, and salt water and soap equal to or better than the standard sample or equal to or better than a rating of "good." The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating not lower than 3.5. Testing shall be as specified in 4.4.3.

3.4.3 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4.3. When no standard sample is available, the finished cloth shall show no more than a slight trace of labile sulfur when tested as specified in 4.4.3.

3.5 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.3.

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TABLE I. Physical requirements

Weight, ounces per square yard (minimum)	Yarns per inch (minimum)		Breaking strength pounds (minimum)	
	Warp	Filling	Warp	Filling
4.5	31	30	50	50

3.5.1 Weave. The weave shall be plain.

3.5.2 Width. The minimum width exclusive of selvages shall be 37-1/2 inches for the Military services and 35-3/4 inches for Civil agencies (see 6.2).

3.6 Finish. The cloth shall be mercerized.

3.7 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material shall not exceed 2.0 percent when tested as specified in 4.4.3.

3.8 pH. The pH value of the water extract of the dyed and finished cloth shall be not less than 5.0 and not greater than 8.5 when tested as specified in 4.4.3.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous pieces of not less than 35 yards and shall be put-up in rolls or flat folded-pieces (bolts) as specified in 5.1.

3.10 Fiber identification. Each roll or folded piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.11 Workmanship. The cloth shall conform to the quality of product established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the applicable established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1 1 Responsibility for compliance All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 through 4.4.2.4 and tested for the characteristics specified in 4.4.3.

4.4 Quality conformance inspection.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination.

4.4.2.1 Yard-by-yard examination. Each roll or flat-folded piece in the sample shall be examined on one side only. When the total yardage in the roll or flat-folded piece does not exceed 100 yards, the entire yardage in the roll or flat-folded piece shall be examined. When the total yardage in the roll or flat-folded piece exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section I of FED-STD-4, which are clearly visible at normal inspection distance (3 feet), shall be scored and assigned demerit points as listed in 4.4.2 1.2 except that only those slubs and knots which exceed the limits shown on figure 1 of FED-STD-4 shall be scored, and coarse yarns and jerked-in filling, glossary numbers 2, 3, and 26 shall not be scored. No linear yard (folded yards, or increments of 1 yard on the measuring device of the inspection machine) from any one piece within the sample shall be penalized more than four points. The sample size shall be 20 rolls, or flat-folded pieces, selected from 20 containers. The lot shall be unacceptable if the points per 100 yards or the total yardage examined exceeds 55 points. The lot shall be

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unacceptable if the points per 100 square yards of two or more individual rolls or folded pieces exceeds 83 points. If one roll or flat-folded piece exceeds 83 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls or flat-folded pieces in the second sample exceeds 83 points per 100 square yards. Point computation for lot quality and individual roll or piece quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.4.2.1.1 Small sized lot yard-by-yard examination. For small purchases, when lot sizes of less than 3200 yards may be presented (see 6.2), the sample unit shall be in yards, randomly selected from the rolls or flat-folded pieces in the lot and the sample yardage shall be examined on one side only. The lot shall be unacceptable if the maximum points per sample are exceeded or if the sample yardage from any individual roll or flat-folded pieces contains greater than 55 points per 100 square yards (using formula cited in 4.4.2.1). The sampling plan shall be as follows:

Lot size (yards)	Sample size (yards)	Maximum points per sample
0-1200	300	165
1201-3199	500	275

4 4 2.1.2 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches ir any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Width less than specified
Overall uncleanness
Poor dye penetration, mottled, streaky, or cloudy

4 4.2.2 Length examination. During the yard-by-yard examination, each roll or flat-folded piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls or flat-folded pieces in the sample are defective with respect to length or if the total of the actual lengths of rolls or flat-folded pieces in the sample is less than the total of the lengths marked on the tickets.

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4.4.2.3 Shade examination. During the yard-by-yard examinations, each roll or flat-folded piece in the sample shall be examined for shade. Any roll or flat-folded piece in the sample off shade or shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Marking examination. During the yard-by-yard examination, each roll or flat-folded piece shall be examined for marking. The lot shall be unacceptable if two or more rolls or flat-folded pieces in the sample are not labeled, ticketed, or invoiced in accordance with the Textile Fiber Products Identification Act.

4.4.3 End item testing. The cloth shall be tested for the characteristics listed in table II. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit for test purposes shall be 1-1/2 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Identification of cotton	3.3.1	1200 1/
Yarn ply	3.3.1	Visual 2/
Colorfastness to:		
Crocking	3.4.2	5651
Light	3.4.2	5660
Water	3.4.2	5630
Salt water and soap	3.4.2	5632
Presence of labile sulfur	3.4.3	2020
Weight	3.5	5041
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050
Breaking strength:		
Warp	3.5	5100
Filling	3.5	5100

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TABLE II. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method
Weave	3.5.1	Visual 2/
Mercerization	3.6	Microscopic examination for swelling of cotton fibers 1/
Nonfibrous material	3.7	2611
pH	3.8	2811

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made from each sample unit and results reported as "pass" or "fail."

4.4.4 Packaging inpsection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1134.

5 PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking specified in the contract or purchase order, shipments shall be marked in accordance with PPP-P-1134.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bunting is primarily used in the manufacture of signal, semaphore, ensign (including U.S. flag), and other types of flags.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. When first article is required (see 3.1, 4.3, and 6.3).
c. Color required (see 3.4).
d Width required (see 3.5.2).
e. Length required if other than specified (see 3.9).
f. When small size lot examination criteria is applicable (see 4.4.2.1.1).
g. Levels of put-up, preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6 5 Dyestuff formulations. The following are suggested dyestuff formulations for some Red, Yellow, Blue, and Black colors:

Old Glory Red 70180

Azoic Coupling 2, C.I. 37505
Azoic Coupling 29, C.I. 37527
Azoic Diazo 9, D.I. 37040

Old Glory Blue 70075

Vat Blue 14, C.I. 69810
Vat Violet 13, C.I. 68700
Vat Violet 9, C.I. 60005
Vat Blue 20, C.I. 59800
Vat Black 27, C.I. 69005

Yellow 65002

Vat Yellow 22 No C I. number

Black 65018

Vat Black 9, C.I. 65230
Vat Black 25, C.I. 69525

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6.6 For civil agency procurement. When level B packaging is required for civil agency procurement, the level A packaging requirements shall apply.

6.7 Subject term (key word) listing.

Flags
Textile Fabrics

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - IH, MD
Air Force - 11, 82
DLA - CT

User activity:

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

COM - NIST
GSA - FSS
JUS - FPI
VA - OSS

PREPARING ACTIVITY:

Army - GL

(Project 8305-0313)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	CCC-C-439G	90/05/15

3. DOCUMENT TITLE

CLOTH, BUNTING, COTTON, MERCERIZED

4 NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.*)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S Army Natick RD&E Center	508-651-5221	256-5221
c ADDRESS (Include Zip Code) Commander, U S Army Natick RD&E Center ATTN STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT. Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041 3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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